Welcome to the course

Ostbayerische Technische Hochschule
Amberg-Weiden

AI Project

Prof. Dr.-Ing. Thomas Nierhoff

AI Project

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The course "AI Project" is only offered in winter semester

- Grades can only be submitted in winter semester
- Even if you conduct your project with another professor in the summer semester, you have to wait till the end of the winter semester for your grade to be submitted



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Goal of the course

- To implement a larger project in the context of AI under supervision
  - as a group of two persons
  - without the hazzle of writing a master thesis

Structure of the course

- Group / topic selection and hardware handout at the beginning of the semester
- Biweekly meetings with every group to discuss the progress of your project
- (Live) presentation at the end of the semester

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Three ways to select a project

- You select a project shown on the next slides
  (at least two teams per project or the project will be discarded)
- You already have a project from another professor
  (then he/she will supervise you and submit your grade at the end of the semester)
- You and 1+ other group(s) want to do a specific project
  - it must not have been submitted in a similar version in another course
  - it should be novel
  - it should involve hardware and/or a "real life" problem

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Deadlines/important dates this semester

- **12.10.26 (11:15 AM)**
  - group selection via https://moodle.oth-aw.de/mod/choicegroup/view.php?id=199032
  - project selection via https://docs.google.com/forms/d/e/1FAIpQLSdFPMQWAnrET6-8EZyCYpFyksQkVXa0Z9nYLxLI3sW059nlXw/viewform
  - You can contact me anytime via email or during the lecture if you have questions

- **19.10.26 (8:00 – 10:30 AM)**
  Hardware handout in my office

- **10.1.27 (23:59 PM)**
  Submission deadline

- **11.1.27**
  Presentation

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Project 1: Water in a bottle

- 3-5 different bottles are filled with predefined amounts of water. You develop a model that is able to predict how much water is in each bottle



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Project 2: Gesture control for the racing simulator TORCS

- You develop a system that is able to control the TORCS racing simulator by holding a virtual steering wheel and pushing virtual pedals with your feet
- One or two webcams are used to detect your gestures



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Project 3: Data analytics for potential new study programs

- Our faculty is always seeking for opportunities to attract new students.
- By analyzing data from different sources (e.g. data from the Federal Statistical Office, scraping other university websites, …) you have to figure out and argue which potential new study program might attract most students and why it is well suited for our faculty



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Project 4: Pokemon

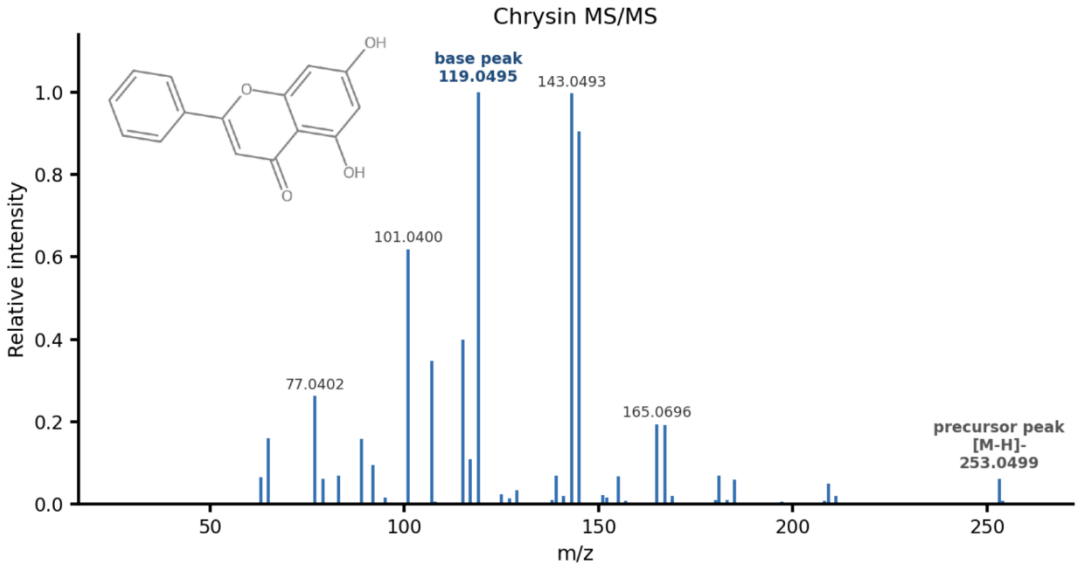
- Participate in the Kaggle "Pokémon Trading Card Game AI Battle Challenge"
- For details see https://www.kaggle.com/competitions/the-pokemon-company-ptcg-ai-battle-challenge-playground



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Project 5: Molecude ID from Mass Spectra

- Participate in the Kaggle "Enveda CASMI 2026 - Molecule ID From Mass Spectra" challenge
- For details see https://www.kaggle.com/competitions/enveda-CASMI26-molecule-id-mass-spectra/overview



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Project 6 (max. 5 teams): Imitation learning using a Niryo robot

- Learn how to push an object along a designated path (black line on the table) towards a goal position using imitation learning



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Project 7 (max. 5 teams, experimental): Loop engineering with HIL (hardware in the loop)

- Develop an agentic workflow for the Turtlebot 4 robot sucht that it is able to push a box towards a specific position
- Note: The goal is to iteratively improve the Turtlebot's code (not a neural network) through an agentic workflow with (ideally) no human intervention

